# Linear Programming Approach to the Graph Square Problem

Edward Xu

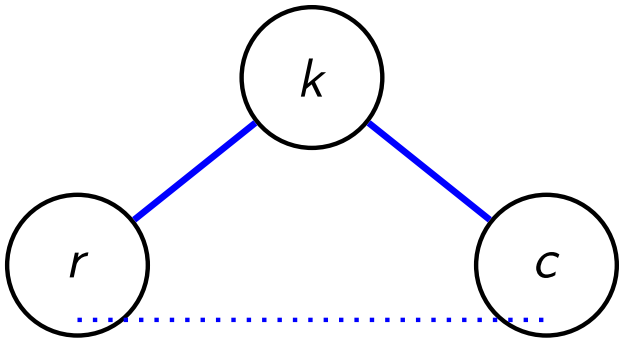
University of Melbourne

Supervisor: Dr Christopher Duffy

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# Square of a Graph

**Two-step connections**

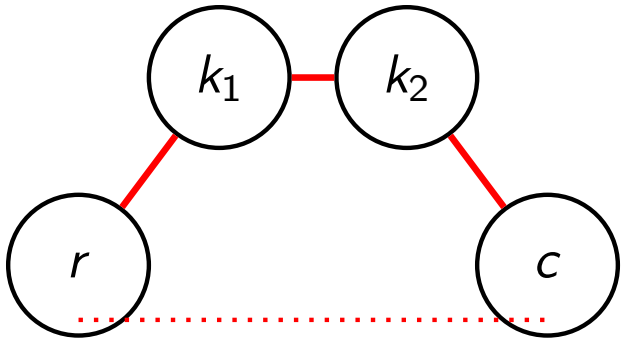



two steps

$$(A^2)_{rc} = \sum_{k=1}^{n} A_{rk} A_{kc}.$$

Counts walks of length two from $r$ to $c$.

**Three-step connections**

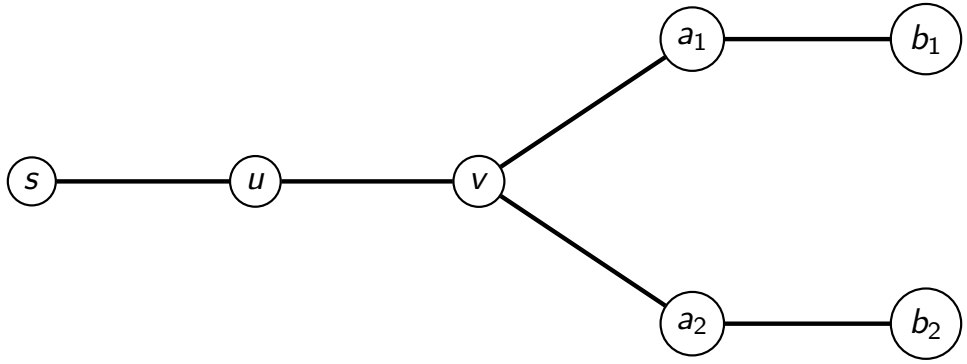



three steps

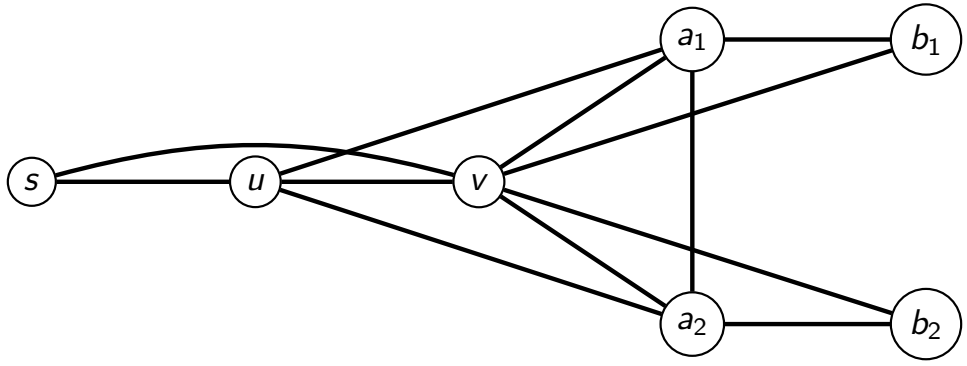
$$(A^3)_{rc} = \sum_{k=1}^{n} A_{rk} (A^2)_{kc}.$$

Counts walks of length three from $r$ to $c$.

## Graph Root




## Graph Square or Square of a Graph



# The Graph Square Problem

## Problem (SQUARE)

**Instance:** *A finite, simple, undirected graph* $G = (V, E)$.

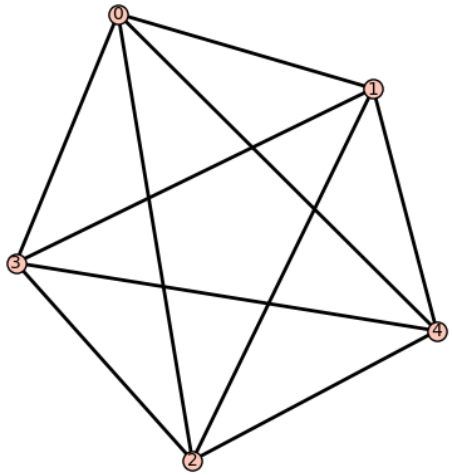
**Question:** *Does there exist a finite, simple, undirected graph* $H$ *such that*

$$G = H^2?$$

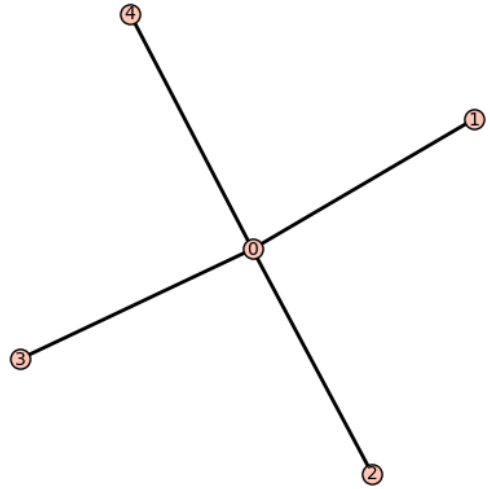
Let the **graph root** $H = (V, E_H)$ be a finite, simple, undirected graph, where $V$ is the vertex set with $|V| = n$, and $E_H \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$ is the edge set. The **square of a graph** $H$, denoted by $H^2 = (V, E_{H^2})$ is the graph on the same vertex set with edge set

$$E_{H^2} = \{uv \mid u \neq v, 1 \leq d_H(u, v) \leq 2\}. \tag{1}$$

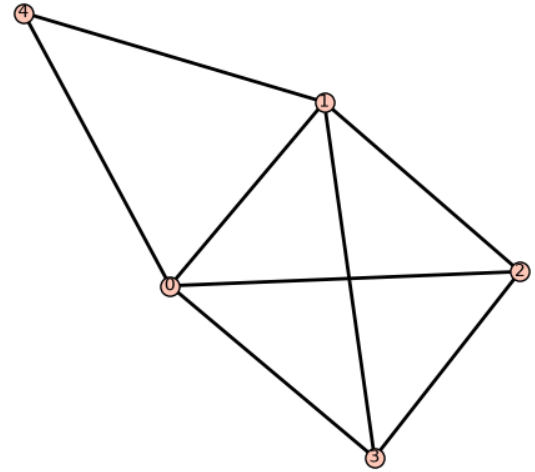
When $G = H^2$ we say that $H$ is a **root** of $G$.



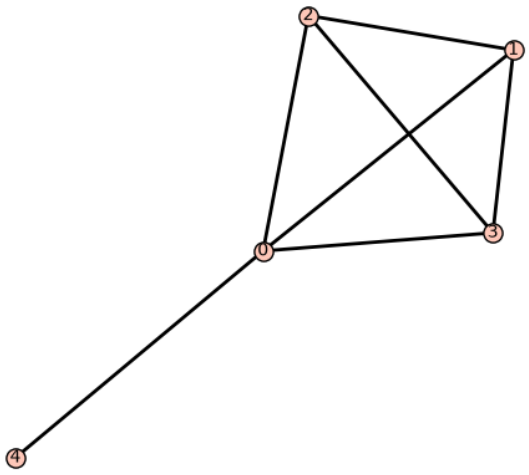

(a) $K_5$, which is a root of itself.



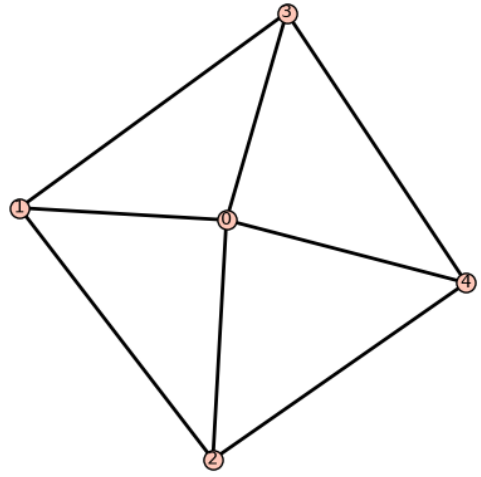

(b) $K_{1,4}$, a root of $K_5$.



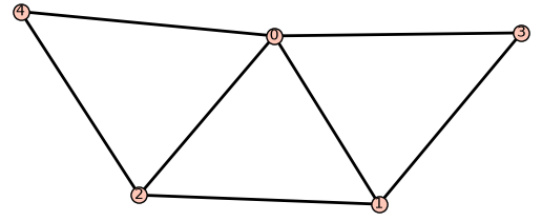

(c) A root of $K_5$.




(d) A root of $K_5$.

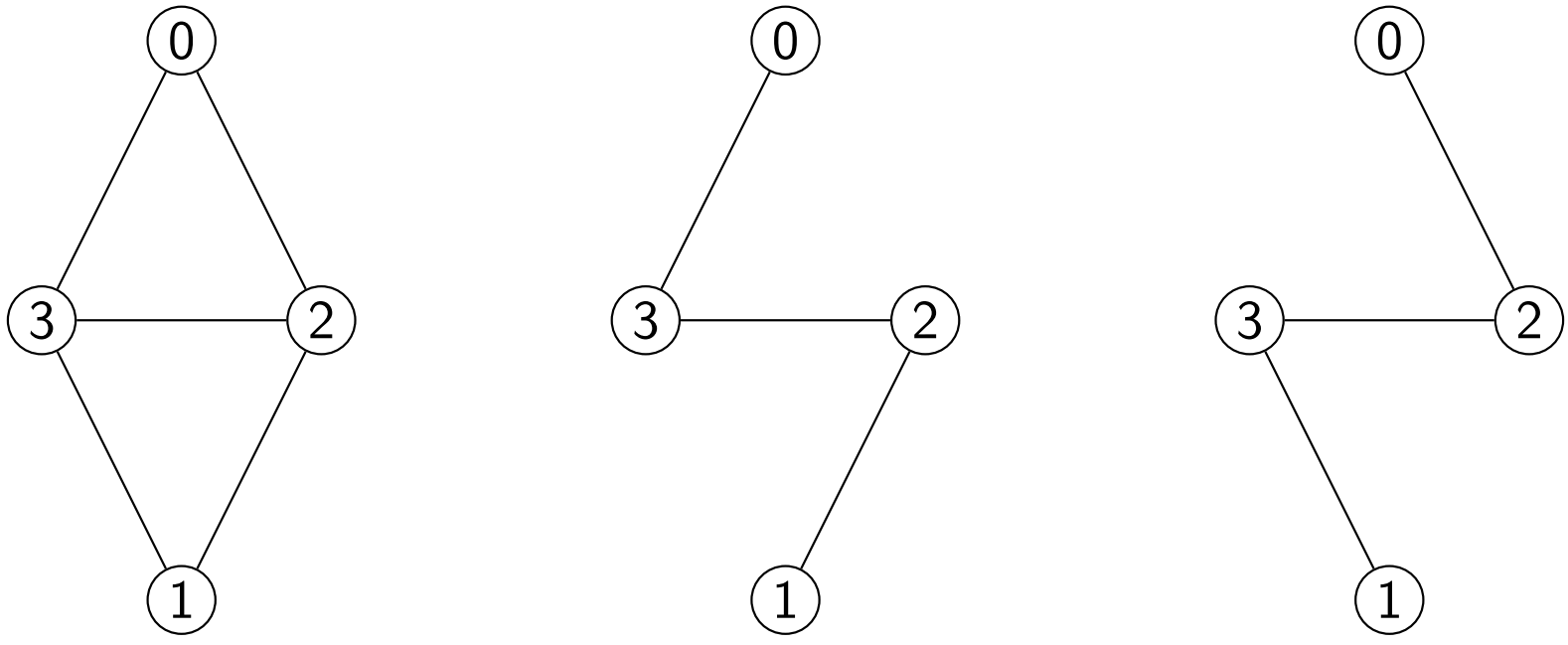



(e) A root of $K_6$.




(f) A root of $K_7$.

Figure: A few of the 15 non-isomorphic graph roots of $K_5$.




(a) $G$ (b) $T_1$ (c) $T_2$

Figure: The graph $G$ has two possible $P_4$ square roots that are trees, $T_1$ and $T_2$. The two roots are isomorphic, differing only by the interchange of vertices 2 and 3.

# NP-Completeness

Motwani and Sudan (1994) proved that the graph square problem is **NP**-complete by showing that the problem is **NP**-hard through a polynomial reduction from Not-All-Equal 3-SAT.

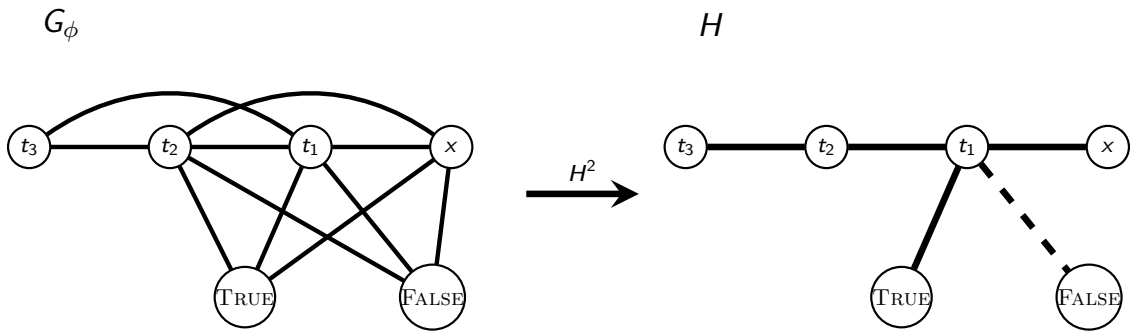
## Problem (NAESAT)

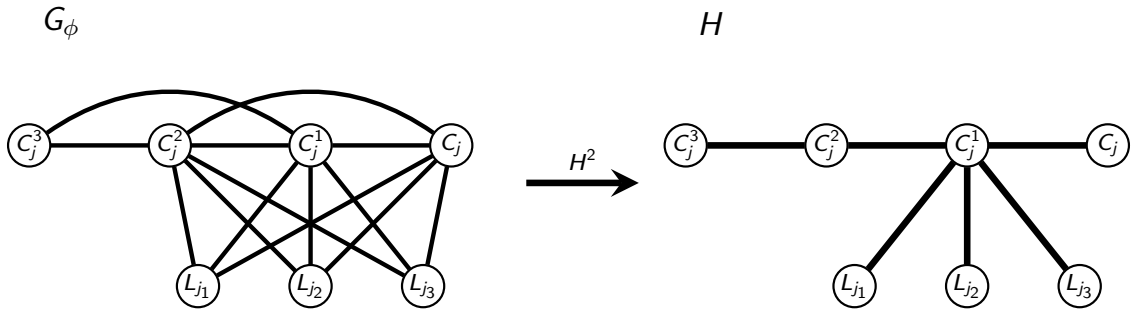
**Instance:** *A 3-CNF formula*

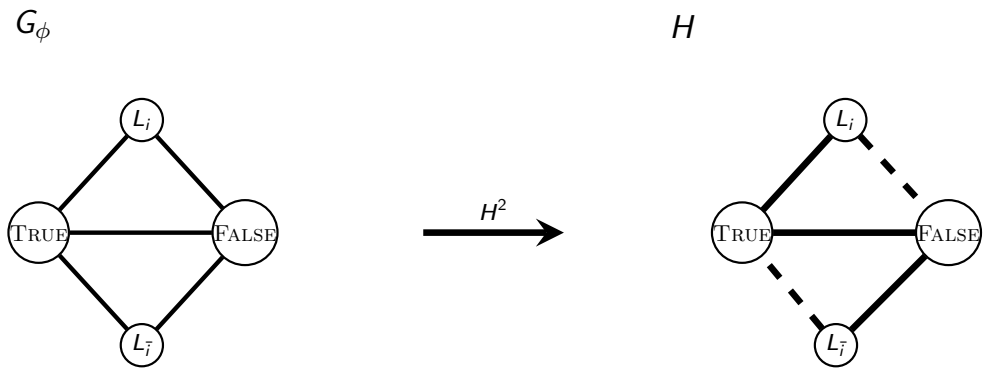
$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$$

*where each clause $C_i$ is over variables $x_1, \ldots, x_n$.*

**Question:** *Does there exist an assignment of* TRUE *and* FALSE *to the variables such that every clause contains at least one* TRUE *literal and at least one* FALSE *literal?*


$G_\phi$
$H$
$t_3$
$t_2$
$t_1$
$x$
TRUE
FALSE
$H^2$


$G_\phi$
$H$
$C_j^3$
$C_j^2$
$C_j^1$
$C_j$
$L_{j_1}$
$L_{j_2}$
$L_{j_3}$
$H^2$


$G_\phi$
$H$
$L_i$
TRUE
FALSE
$L_{\bar{i}}$
$H^2$

# Restricting the Square vs the Root

| **Square $G$** | **Root $H$** | **Main result** |
|---|---|---|
| Unrestricted | Trees | Ross and Harary (1960): structural characterization and reconstruction of tree roots, unique up to isomorphism [RH60]. |
| Unrestricted | Trees | Lin and Skiena (1995): optimal linear-time recognition and reconstruction of tree roots [LS95]. |
| 2-connected | Unrestricted | Fleischner (1974): the square of a 2-connected graph has a Hamiltonian cycle [Fle74]. |
| Unrestricted | Interval squares | Lau and Corneil (2004): polynomial-time construction of roots from interval squares [LC04]. |
| Unrestricted | Block graphs | Le and Tuy (2007): linear-time recognition of roots of block graphs [LT07]. |
| Unrestricted | Bipartite | Lau (2006): linear-time construction of bipartite roots [Lau06]. |
| Directed graphs | Directed graphs | Geller (1970): necessary and sufficient conditions for directed square-root existence [Gel70]. |
| Unrestricted | Graphs of prescribed girth | Farzad, Lau, Le, and Tuy (2009) and Farzad and Karimi (2012): complexity dichotomy for square-root recognition [Far+09; FK12]. |
| Maximum degree $\leq 6$ | Unrestricted | Cochefert, Couturier, Golovach, Kratsch, and Paulusma (2013): minimum-edge roots in polynomial time and maximal-edge roots in exponential time [Coc+13]. |
| Unrestricted | Oriented graphs, maximum degree $\leq 3$, girth $\geq 4$ | Duffy (2025): classification of squares of oriented graphs under these root restrictions [Duf25]. |

# Maximal Cliques and Unicliqual Vertices in Tree Roots

**Input:** A graph $G$ where $G = T^2$
**Output:** A tree $T$
**Step 1**
**if** $G = K_p$ is complete **then**
    **Return** tree $T$ with depth 1, root any vertex
**else**
    **Continue**
**end if**
**Step 2**
Identify all maximal cliques $C_1, \ldots, C_n$ of $G$ (e.g. Bron–Kerbosch [BK73])
**Step 3**
Let $b_1, \ldots, b_n$ be multicliqual points for $C_1, \ldots, C_n$
Build graph $T$ on $\{b_1, \ldots, b_n\}$
**for** each pair $(C_i, C_j)$ with $|C_i \cap C_j| = 2$ **do**
    Add edge $(b_i, b_j)$ in $T$
**end for**
$T$ is a tree
**Step 4**
**for** each clique $C_i$ **do**
    Let $n_i$ be the number of unicliqual points in $C_i$
    Attach $n_i$ leaves to $b_i$ in $T$
**end for**
Resulting $T$ has $|V(G)|$ nodes and $T^2 = G$

# Integer Program

Two distinct vertices are adjacent in $G = H^2$ if and only if they are connected by a path of length one or two in $H$.

$$A_G = (A_H)^2 \vee A_H. \tag{2}$$

$$(A_H)^2_{ij} = \big((A_H)_{i1} \wedge (A_H)_{1j}\big) \vee \big((A_H)_{i2} \wedge (A_H)_{2j}\big) \vee \cdots \vee \big((A_H)_{in} \wedge (A_H)_{nj}\big)$$

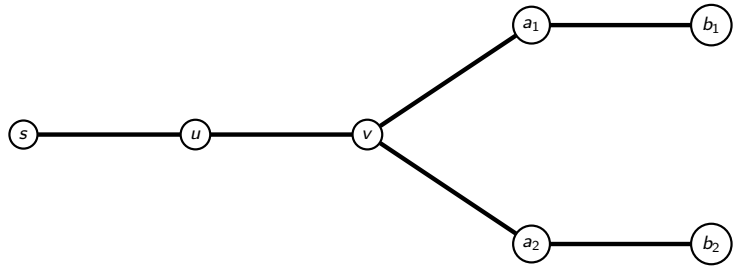
$$g_{ij} \text{ if and only if } \Big(\bigwedge_k (x_{ik} \wedge x_{kj}) \vee x_{ij}\Big). \tag{3}$$

## Lemma

*Let $G$ and $H$ be graphs with adjacency matrices $A_G$ and $A_H$,respectively. The graph $H$ is a root of $G$ if and only if*

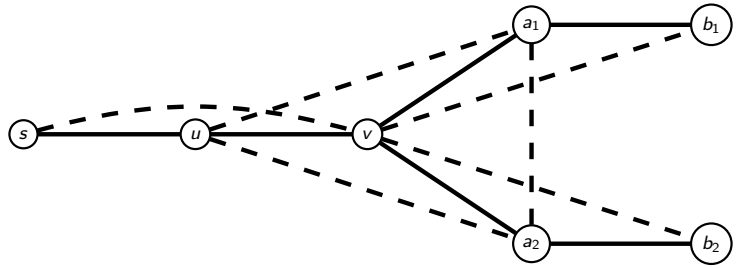
$$A_G = \mathbb{1}_{\geq 1}\big((A_H)^2 - D_H + A_H\big).$$

Vertices ordered $s, u, v, a_1, a_2, b_1, b_2$.




$$A_H = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix} \qquad (A_H)^2 = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 2 & 2 & 0 & 0 \\ 1 & 0 & 3 & 0 & 0 & 2 & 2 \\ 0 & 2 & 0 & 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$D_H = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$




$$A_H^2 - D_H + A_H = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 2 & 2 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 0 & 1 & 1 & 0 \\ 0 & 2 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix} \mathbb{1}_{\geq 1}\left(A_H^2 - D_H + A_H\right) = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}$$

$$A_G = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}$$

i) $x_{ij} \in \{0,1\}$ denotes the $(i,j)$ entry of the adjacency matrix $A_H$.

ii) $y_{ikj} \in \{0,1\}$ represents the product $x_{ik}x_{kj}$ in $(A_H)^2_{ij}$.

iii) $z_{ij} \in \mathbb{Z}_{\geq 0}$ represents the argument of the indicator function.

iv) $M \in \mathbb{Z}_{>0}$ be a sufficiently large constant $M \leq n(n-1) = O(n^2)$.

$$D_H = \begin{bmatrix} \deg_H(1) & 0 & \cdots & 0 & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ \vdots & \cdots & \deg_H(i) & \cdots & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & \deg_H(n) \end{bmatrix}, \quad A_H = \begin{bmatrix} x_{11} & \cdots & x_{1j} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & \cdots & x_{ij} & \cdots & x_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nj} & \cdots & x_{nn} \end{bmatrix}$$

$$(A_H)^2 = \begin{bmatrix} \sum_k x_{1k}x_{k1} & \cdots & \sum_k x_{1k}x_{kj} & \cdots & \sum_k x_{1k}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_k x_{ik}x_{k1} & \cdots & \sum_k x_{ik}x_{kj} & \cdots & \sum_k x_{ik}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_k x_{nk}x_{k1} & \cdots & \sum_k x_{nk}x_{kj} & \cdots & \sum_k x_{nk}x_{kn} \end{bmatrix}.$$

$$\begin{bmatrix} g_{11} \\ \vdots \\ g_{1j} \\ \vdots \\ g_{1n} \\ \vdots \\ g_{ij} \\ \vdots \\ g_{nn} \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}(z_{11}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1j}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1n}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{ij}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{nn}) \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{1k}x_{k1} - (D_H)_{11} + x_{11}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{1k}x_{kj} - (D_H)_{1j} + x_{1j}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{1k}x_{kn} - (D_H)_{1n} + x_{1n}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{ik}x_{kj} - (D_H)_{ij} + x_{ij}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{nk}x_{kn} - (D_H)_{nn} + x_{nn}\big) \end{bmatrix}. \tag{4}$$

Objective Function:

$$\max \quad 0 \tag{5}$$

Variables:

$$x_{ij} \in \mathbb{Z}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{6}$$

$$y_{ikj} \in \mathbb{Z}_{\geq 0}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{7}$$

$$z_{ij} \in \mathbb{Z}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{8}$$

Constraints:

(1) Indicator argument:

$$z_{ii} = \sum_{k=1}^{n} y_{iki} - \sum_{k=1}^{n} x_{ik}, \forall i \in \{1, \ldots, n\}, \tag{9}$$

$$z_{ij} = \sum_{k=1}^{n} y_{ikj} + x_{ij}, \qquad \forall i \neq j \in \{1, \ldots, n\}, \tag{10}$$

(2) Indicator function:

$$z_{ij} \leq M g_{ij}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{11}$$

$$z_{ij} \geq g_{ij}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{12}$$

(3) Product:

$$y_{ikj} \leq x_{ik}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{13}$$

$$y_{ikj} \leq x_{kj}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{14}$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{15}$$

(4) Upper Bound:

$$x_{ij} \leq 1, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{16}$$

$$y_{ikj} \leq 1, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{17}$$

(5) Symmetry:

$$x_{ij} = x_{ji}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{18}$$

(6) Diagonal:

$$x_{ii} = 0, \qquad \forall i \in \{1, \ldots, n\}. \tag{19}$$

The integer program on the arbitrary graph $G$ using Equations (6)–(19) together with the constant objective function Equation (5) is denoted as $\mathcal{IP}(G)$.
The total number of simple graphs on $n = 2, 3, 4, \ldots$ unlabeled nodes is the sequence $2, 4, 11, 34, 156, 1044, 12346, \ldots$ given by the OEIS sequence A000088. The number of graphs which are squares is the sequence $2, 3, 6, 11, 28, 77, 307, \ldots$ given by A382181.

| $n$ | $H$ exists | $H$ does not exist |
|---|---|---|
| 3 | 3 | 1 |
| 4 | 6 | 5 |
| 5 | 11 | 23 |
| 6 | 28 | 128 |
| 7 | 77 | 967 |
| 8 | 307 | 12039 |

Table: Outcomes of the integer program for graphs of different orders of $n$.

# Methodology

Implemented in **SageMath**, with **Gurobi** as the optimization backend.

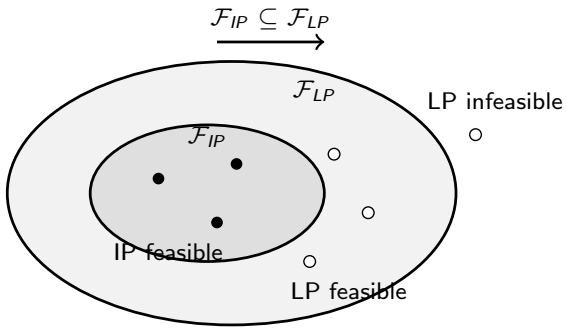
Uses the **barrier method**, an interior-point method based on predictor-corrector Newton iterations.

Results correspond to the labelled representatives produced by the specified generation procedure.

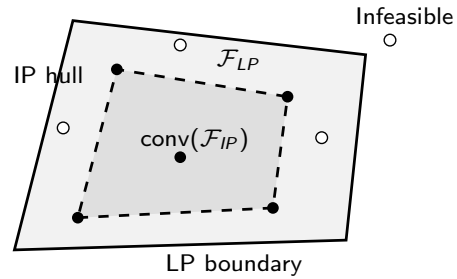
Code and test harness: `https://github.com/Hellwoodoo/Graph-Squares`.

```
def linear_program(G):
    m = MixedIntegerLinearProgram(solver='Gurobi', maximization=False)
    m.solver_parameter("Method", 2)
    x = m.new_variable(real=True, nonnegative=True)
    y = m.new_variable(real=True, nonnegative=True)
    z = m.new_variable(real=True, nonnegative=True)
    A_G = G.adjacency_matrix()
    n = A_G.nrows()
    M = n**2
    for i in range(n):
        for j in range(n):
            if i == j:
                m.add_constraint(z[i,j] - (sum(y[i,k,j] for k in range(n)) - sum(x[i,k] for k in range(n)) + x[
    i,j]) == 0)
            else:
                m.add_constraint(z[i,j] - (sum(y[i,k,j] for k in range(n)) + x[i,j]) == 0)
    for i in range(n):
        for j in range(n):
            m.add_constraint(z[i,j] - M * A_G[i][j] <= 0)
            m.add_constraint(-z[i,j] + A_G[i][j] <= 0)
    for i in range(n):
        for j in range(n):
            for k in range(n):
                m.add_constraint(y[i,k,j] - x[i,k] <= 0)
                m.add_constraint(y[i,k,j] - x[k,j] <= 0)
                m.add_constraint(- y[i,k,j] + x[i,k] + x[k,j] - 1 <= 0)
    for i in range(n):
        for j in range(n):
            m.add_constraint(x[i,j] == x[j,i])
    for i in range(n):
        m.add_constraint(x[i,i] == 0)
    for i in range(n):
        for j in range(n):
            m.add_constraint(x[i,j] - 1 <= 0)
            for k in range(n):
                m.add_constraint(y[i,k,j] - 1 <= 0)
    return m, x
```

# Linear Relaxation




**Set-theoretic**




**Geometric**

$$x_{ij} \in \mathbb{R}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{20}$$

$$y_{ikj} \in \mathbb{R}_{\geq 0}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{21}$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}. \tag{22}$$

Such a linear program may have *integral solutions* and *fractional solutions* when the root exists and *fractional solutions* and *infeasible solutions* when the root does not exist.

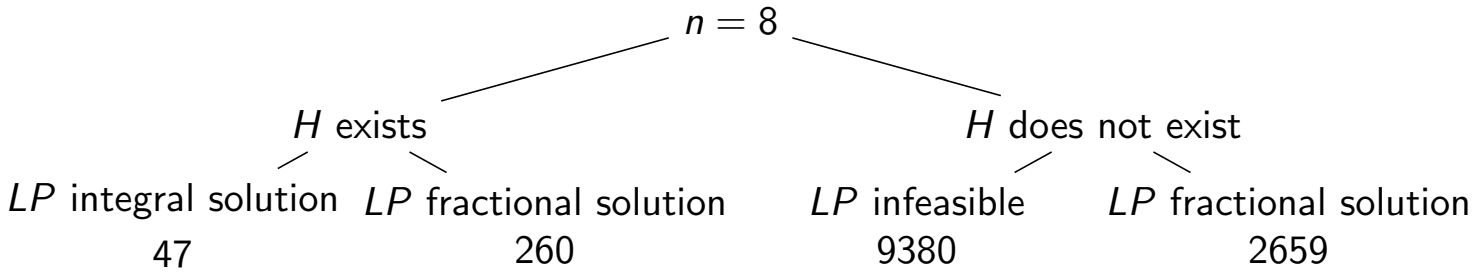



Figure: $LP$ outcomes for $n = 8$ vertices using $\mathcal{LP}(G)$.

The $LP$ integral solution sequence $2, 3, 6, 10, 17, 29, 51, \ldots$ matches A285665. However, it is unclear why this should be the case.

| $n$ | $H$ exists | | $H$ does not exist | |
|---|---|---|---|---|
| | $LP$ integral solution | $LP$ fractional solution | $LP$ infeasible | $LP$ fractional solution |
| 3 | 3 | 0 | 1 | 0 |
| 4 | 5 | 1 | 5 | 0 |
| 5 | 7 | 4 | 20 | 3 |
| 6 | 13 | 15 | 107 | 21 |
| 7 | 22 | 55 | 767 | 200 |
| 8 | 44 | 263 | 9380 | 2659 |

Table: Outcomes of the $LP$ relaxation for graphs of different orders $n$, classified according to whether a square root $H$ exists.

The linear relaxation will never return an infeasible solution if $H$ exists.

## Proposition

*If $G$ has a square root $H$, then $\mathcal{LP}(G)$ is feasible.*

## Proof.

Since $\mathcal{LP}(G)$ is obtained by relaxing the integrality constraints of $\mathcal{IP}(G)$, every feasible solution of $\mathcal{IP}(G)$ is also feasible for $\mathcal{LP}(G)$. Hence $\mathcal{LP}(G)$ is feasible. □

# Min and Max Objective Functions

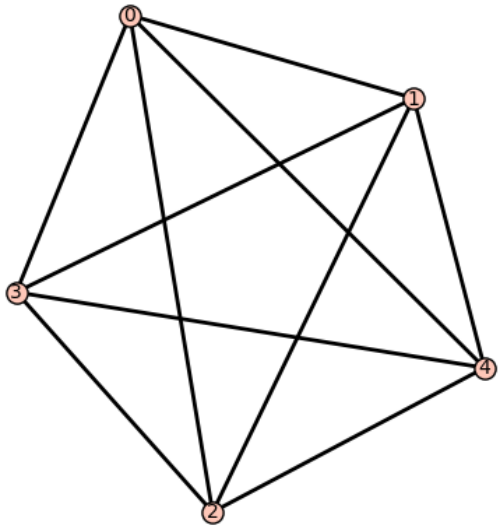
$$\min \sum_{i=1}^{n} \sum_{j=1}^{n} x_{ij}$$

| $n$ | $H$ exists | | $H$ does not exist | |
|---|---|---|---|---|
| | LP integral solution | LP fractional solution | LP infeasible | LP fractional solution |
| 3 | 2 | 1 | 1 | 0 |
| 4 | 3 | 3 | 5 | 0 |
| 5 | 3 | 8 | 20 | 3 |
| 6 | 4 | 24 | 107 | 21 |
| 7 | 5 | 72 | 767 | 200 |
| 8 | 7 | 300 | 9380 | 2659 |

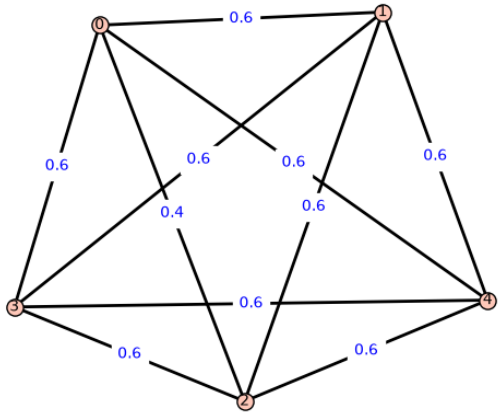
$$\max \sum_{i=1}^{n} \sum_{j=1}^{n} x_{ij}$$

| $n$ | $H$ exists | | $H$ does not exist | |
|---|---|---|---|---|
| | LP integral solution | LP fractional solution | LP infeasible | LP fractional solution |
| 3 | 3 | 0 | 1 | 0 |
| 4 | 6 | 0 | 5 | 0 |
| 5 | 9 | 2 | 20 | 3 |
| 6 | 18 | 10 | 107 | 21 |
| 7 | 38 | 39 | 767 | 200 |
| 8 | 94 | 213 | 9380 | 2659 |

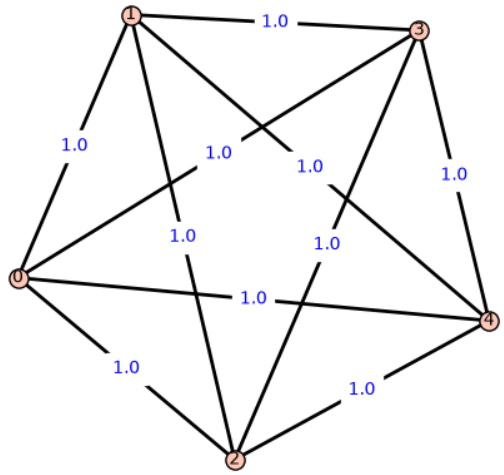
The single complete graph $K_5$ is selected as the unique optimal solution.




(a) $K_5$.




(b) Linear program result for $K_5$.




(c) Max linear program result for $K_5$.

Figure: The graph $K_5$ and two of its square roots.

# Fundamental Properties

## Lemma

*Let $G$ and $H$ be graphs such that $G = H^2$. Let $a, b \in V_G$ such that $ab \notin E_G$. For every feasible solution $x$ of $\mathcal{LP}(G)$ and every $v \in V_G$ with $v \neq a, b$ we have $x_{ab} = y_{avb} = 0$.*

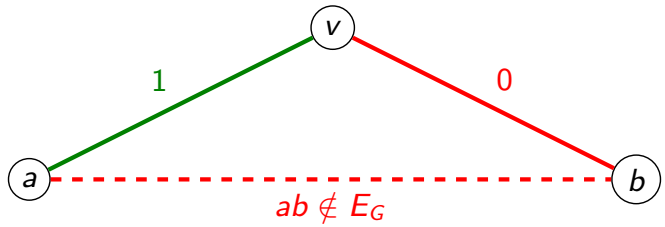
## Lemma

*Let $G$ and $H$ be graphs such that $G = H^2$. Let $a, b \in V_G$ such that $ab \notin E_G$. For every feasible solution $x$ of $\mathcal{LP}(G)$ and every $v \in V_G$ with $v \neq a, b$ we have*

$$x_{av} + x_{vb} \leq 1.$$

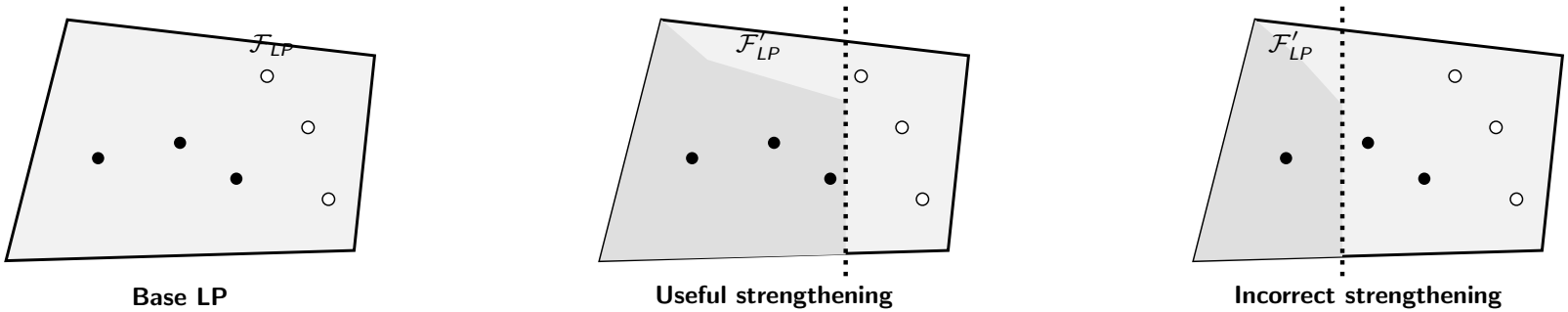
## Lemma

*Let $G$ and $H$ be graphs such that $G = H^2$. Let $a, b, v \in V_G$ such that $ab \notin E_G$. If a feasible solution $x$ of $\mathcal{LP}(G)$ satisfies $x_{av} = 1$, then $x_{vb} = 0$.*



# Strengthening the LP

**Goal:** build a new LP $\max \mathcal{UVLP}(G)$ by adding constraints that let all tree roots be identified.



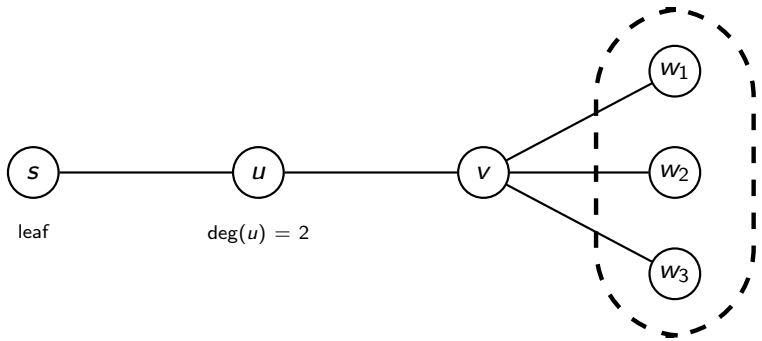

**Base LP** **Useful strengthening** **Incorrect strengthening**

A suitable constraint eliminates fractional solutions while retaining all valid integer solutions, whereas an overly restrictive constraint also excludes valid integer solutions.

# Identifying Square Characteristics of Tree Roots

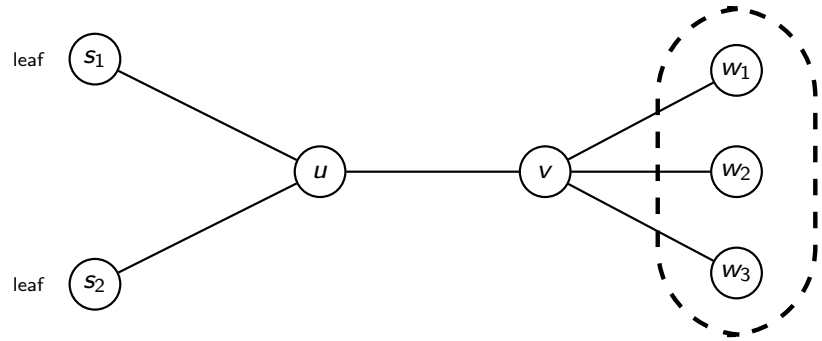
## Definition (Sparse Pendant)

Let $H$ be a graph. A **sparse pendant** is a path $s - u - v$ in $H$ such that $s$ is a leaf and $u$ has degree 2.




## Definition (Pendant Fork)

Let $H$ be a graph. A **pendant fork** is the induced subgraph of $H$ on four vertices $s_1, s_2, u, v$, where $s_1$ and $s_2$ are leaves adjacent to $u$, and every neighbour of $u$ other than $v$ is a leaf.



# Sparse Pendants or Pendant Forks in Trees

## Lemma

*Every non-trivial tree with at least 3 vertices contains either a sparse pendant or a pendant fork subgraph.*

## Proof.

Root $T$ at any vertex. Let $s$ be a leaf of maximum depth, $u$ its parent. Every child of $u$ is a leaf, else a deeper vertex than $s$ would exist. **Case 1: $u$ is not the root.** Let $v$ be the parent of $u$. Every neighbour of $u$ other than $v$ is a leaf child.

If $u$ has exactly one leaf neighbour $s$, then $\deg_T(u) = 2$, and $\{s, u, v\}$ induces a sparse pendant.

If $u$ has two leaf neighbours $s_1, s_2$, then $\{s_1, s_2, u, v\}$ induces a pendant fork.

**Case 2: $u$ is the root.** All neighbours of $u$ are leaf children, and $u$ has at least two of them.

If $u$ has exactly two neighbours $s, v$, then $\{s, u, v\}$ induces a sparse pendant.

Otherwise, pick distinct children $s_1, s_2, v$ of $u$; then $\{s_1, s_2, u, v\}$ induces a pendant fork.

# Leaf Fixing Linear Program

Using Lemma (9) we can identify $s$ to be a leaf, $u$ its corresponding stem and vertex $v$ to give the fix

$$x_{su} = x_{us} = x_{uv} = x_{vu} = 1. \tag{23}$$

The construction of the linear program on the arbitrary graph $G$ using $\max \mathcal{LP}(G)$ with the addition of Equation (**??**) is denoted as $\max \mathcal{LP}(G)$.

## Lemma

*Let $G$ and $T$ be graphs such that $G = T^2$ and $T$ is a tree.* $\max \mathcal{UVLP}(G)$ *always fixes a sparse pendant or pendant fork subgraph of $T$.*

## Proposition (Validity of Unicliqual Vertex Fix)

*If a root $H$ of $G$ exists, then* $\max \mathcal{UVLP}(G)$ *is feasible.*

$\max \mathcal{LP}$

| $n$ | $H$ exists | | $H$ does not exist | |
|---|---|---|---|---|
| | integral | fractional | infeasible | fractional |
| 3 | 3 | 0 | 1 | 0 |
| 4 | 6 | 0 | 5 | 0 |
| 5 | 10 | 1 | 21 | 2 |
| 6 | 22 | 6 | 117 | 11 |
| 7 | 51 | 26 | 852 | 115 |
| 8 | 148 | 159 | 10271 | 1768 |

$\max \mathcal{UVLP}$

| $n$ | $H$ exists | | $H$ does not exist | |
|---|---|---|---|---|
| | integral | fractional | infeasible | fractional |
| 3 | 3 | 0 | 1 | 0 |
| 4 | 6 | 0 | 5 | 0 |
| 5 | 9 | 2 | 20 | 3 |
| 6 | 18 | 10 | 107 | 21 |
| 7 | 38 | 39 | 767 | 200 |
| 8 | 94 | 213 | 9380 | 2659 |

## Theorem

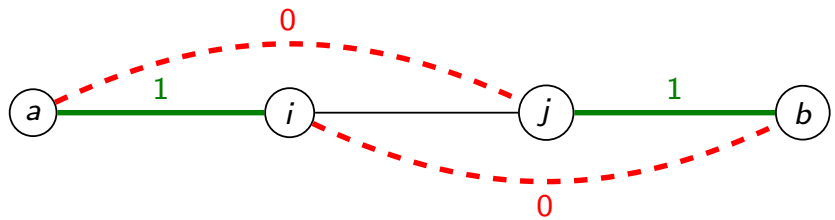
*Let $G$ and $T$ be graphs such that $G = T^2$ and $T$ is a tree. Then $\mathcal{UVLP}(G)$ returns an integral solution.*

# Propagation of the Fixing

## Lemma (0-Path Propagation)

*Let $G$ and $H$ be graphs such that $G = H^2$. Let $a, i, j, b \in V_G$ such that $a, i, j, b$ form a shortest induced path in $H$ between $a$ and $b$. For every feasible solution $x$ of $\mathcal{LP}(G)$,*

$$x_{ai} = 1 \implies x_{ib} = 0, \quad x_{ia} = 1 \implies x_{bi} = 0, \quad x_{jb} = 1 \implies x_{aj} = 0, \quad x_{bj} = 1 \implies x_{ja} = 0.$$



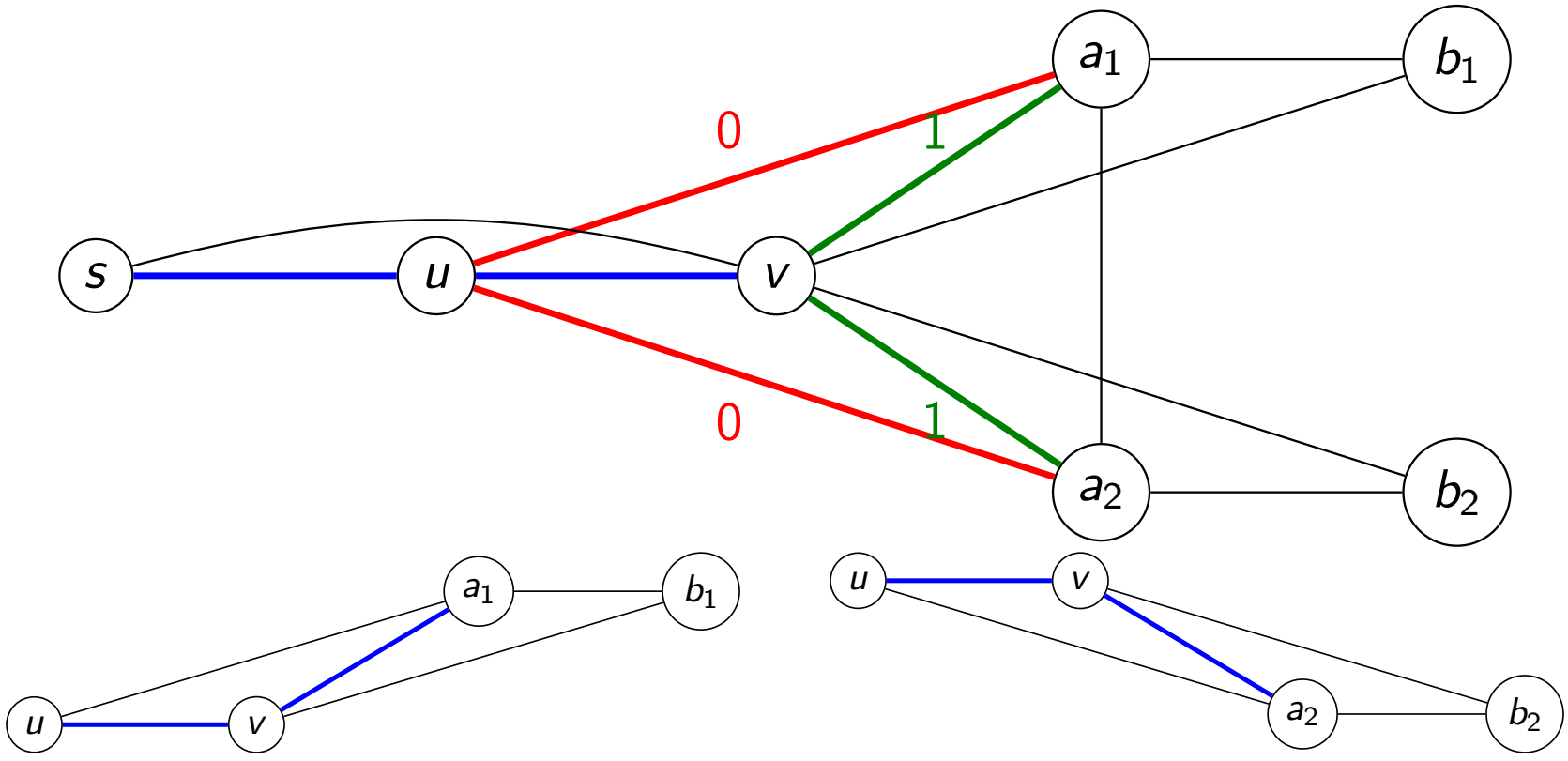

## Lemma (1-Path Propagation)

*Let $G$ be a graph and let $x$ be a feasible solution of $\mathcal{LP}(G)$. Suppose that $u - v - a_1$ form a shortest induced path in $H$ between $u$ and $a_1$ and that $x_{uv} = 1$. Let $A = \{a_1, \ldots, a_r\}$ and $B = \{b_1, \ldots, b_s\}$ be such that every vertex $k \in V_G \setminus \{u, v\}$ satisfies $k \in A \cup B$, and suppose that*

$$x_{um} = 0 \text{ for every } m \in A, \quad g_{un} = 0 \text{ for every } n \in B,$$

*and $g_{sa_1} = 0$ for the vertex $s$ preceding $u$ on the shortest path to $a_1$. Then*

$$x_{va_1} = 1.$$


$a_1$
$b_1$
0
1
$s$
$u$
$v$
0
1
$a_2$
$b_2$
$a_1$
$b_1$
$u$
$v$
$u$
$v$
$a_2$
$b_2$

# Symbolic Sets

## Conjecture

*Let $G$ and $H$ be graphs such that $G = H^2$. Let $a, b \in V_G$ such that $ab \notin E_G$. For every feasible solution $x$ of* $\max \mathcal{LP}(G)$ *and every $v \in V_G$ with $v \neq a, b$ we have*

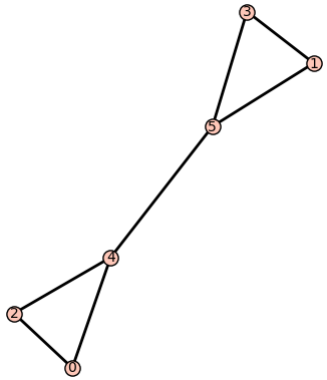
$$x_{av} + x_{vb} = 1 \text{ or } x_{av} + x_{vb} = 0.$$

## Definition (Consistent Set)

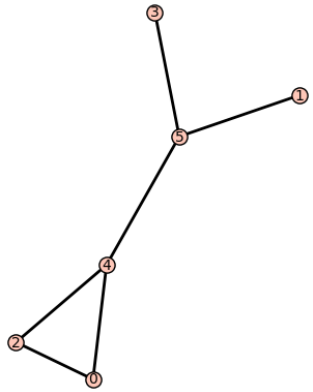
Take a maximal set $E' \subseteq E_G$ of at least three edges in which every edge is linked to another edge of $E'$ by the $= 1$ relationship. We call $E'$ **consistent** if its edges can be split into two nonempty groups $A$ and $B$ so that no two edges within $A$ form a $P_3$ in $G$, and the same holds within $B$.

## Definition (Inconsistent Set)

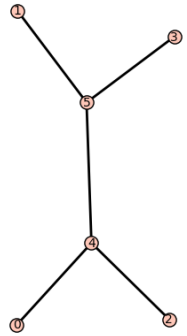
Let $E'$ be as above. We call $E'$ **inconsistent** if no such split exists. That is, if we divide its edges into two nonempty groups $A$ and $B$, at least one of the groups contains two edges forming a $P_3$ in $G$.



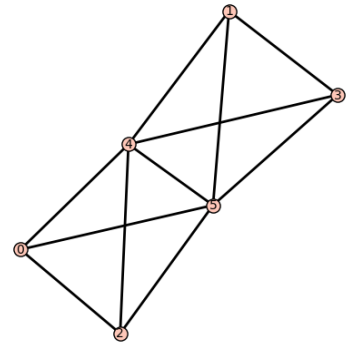

(a) The root graph $G_1$.




(b) The root graph $G_2$.



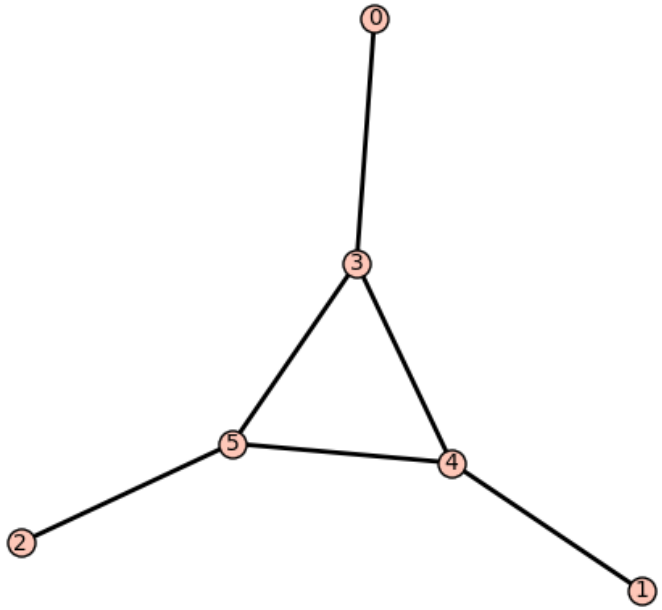

(c) The root graph $G_3$.



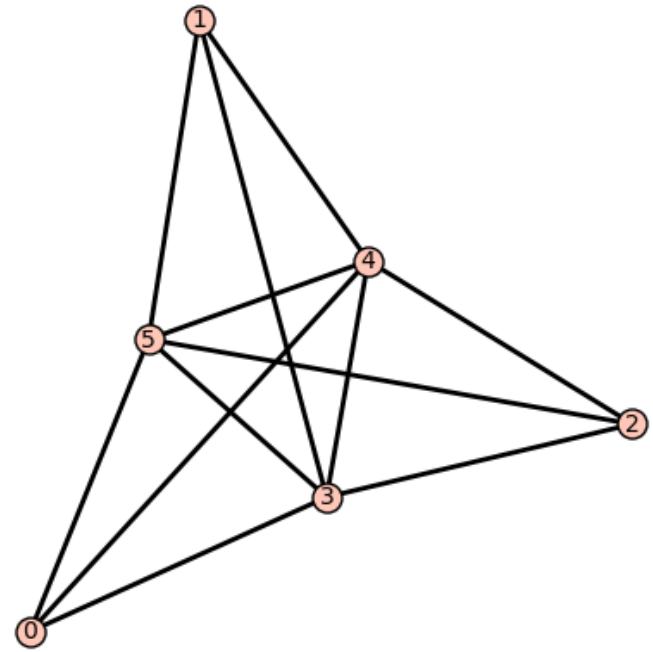

(d) The graph square $H = G_i^2$.

Figure: Three non-isomorphic square roots of the same six-vertex graph square.

$\{(1,4),(4,0),(4,2),(4,3)\}$ and $\{(1,5),(3,5),(5,0),(5,2)\}$ are consistent sets




(a) The root graph $G$.




(b) The graph square $H$ such that $H^2 = G$.

Figure: A six-vertex graph and its square. $G$ only has one root, although there are different relabellings.

$\{(0,3),(3,1),(3,2)\}$, $\{(0,5),(1,5),(5,2)\}$ and $\{(4,2),(4,1),(4,0)\}$ are inconsistent sets

max $\mathcal{LP}$

| $n$ | $H$ exists | | $H$ does not exist | |
|---|---|---|---|---|
| | integral | fractional | infeasible | fractional |
| 3 | 3 | 0 | 1 | 0 |
| 4 | 6 | 0 | 5 | 0 |
| 5 | 10 | 1 | 21 | 2 |
| 6 | 22 | 6 | 117 | 11 |
| 7 | 51 | 26 | 852 | 115 |
| 8 | 148 | 159 | 10271 | 1768 |

**Edge enforcement (base:** max $\mathcal{LP}$**)**

| $n$ | $H$ exists | | $H$ does not exist | |
|---|---|---|---|---|
| | integral | fractional | infeasible | fractional |
| 3 | 3 | 0 | 1 | 0 |
| 4 | 6 | 0 | 5 | 0 |
| 5 | 10 | 1 | 20 | 3 |
| 6 | 26 | 2 | 107 | 21 |
| 7 | 67 | 10 | 767 | 200 |
| 8 | 240 | 67 | 9380 | 2659 |

Outcomes of the greedy iterative edge enforcement linear program using the max linear program as the base for graphs of different orders of $n$ as compared to the original max linear program.

# Rounding

**Input:** A graph $G$
**Output:** A $LP$ model $(m, x)$

Initialize $F = \emptyset$ and construct $\mathcal{UVLP}(G)$.
Solve $\mathcal{UVLP}(G)$.
**while** the current $LP$ solution is fractional **do**
    For each fractional edge $ab$, compute its rounding score $s_{ab} = |x_{ab} - 0.5|$.
    Select a fractional edge $ab$ with maximum score.
    Set

$$\widehat{x}_{ab} = \widehat{x}_{ba} = \begin{cases} 0, & x_{ab} < 0.5, \\ 1, & x_{ab} \geq 0.5. \end{cases}$$

    Add $ab$ with its assigned value to $F$.
    Reconstruct $\mathcal{UVLP}(G)$ with all fixings in $F$.
    Solve $\mathcal{UVLP}(G) + F$.
    **if** $\mathcal{UVLP}(G) + F$ is infeasible **then**
        **return** the previous feasible solution $(m, x)$.
    **end if**
    **if** $\mathcal{UVLP}(G) + F$ is integral **then**
        **return** the integral solution $(m, x)$.
    **end if**
**end while**
**return** $(m, x)$.

| $n$ | $H$ exists | | $H$ does not exist | | Integral $\leftarrow$ Fractional |
|---|---|---|---|---|---|
| | $LP$ integral | $LP$ fractional | $LP$ infeasible | $LP$ fractional | |
| 3 | 3 | 0 | 1 | 0 | 0 |
| 4 | 6 | 0 | 5 | 0 | 0 |
| 5 | 11 | 0 | 20 | 3 | 2 |
| 6 | 27 | 1 | 107 | 21 | 9 |
| 7 | 73 | 4 | 767 | 200 | 35 |
| 8 | 245 | 62 | 9380 | 2659 | 151 |

Table: Outcomes of the greedy rounding linear program using the max linear program as the base for graphs of different orders of $n$.

| $n$ | $H$ exists | | $H$ does not exist | | Integral $\leftarrow$ Fractional |
|---|---|---|---|---|---|
| | $LP$ integral | $LP$ fractional | $LP$ infeasible | $LP$ fractional | |
| 3 | 3 | 0 | 1 | 0 | 0 |
| 4 | 6 | 0 | 5 | 0 | 0 |
| 5 | 11 | 0 | 21 | 2 | 1 |
| 6 | 28 | 0 | 117 | 11 | 6 |
| 7 | 76 | 1 | 852 | 115 | 25 |
| 8 | 268 | 39 | 10271 | 1768 | 120 |

Table: Outcomes of the greedy rounding linear program using the unicliqual vertex linear program as the base for graphs of different orders of $n$.